Vector Fields and Line Integrals

Vector fields and divergence

1 For the vector field $\mathbf{F}(x, y) = \langle x^2y,\ xy^2 \rangle$, find the divergence $\nabla \cdot \mathbf{F}$.

2 For $\mathbf{F}(x, y, z) = \langle y,\ -x, z^2 \rangle$, find the curl $\nabla \times \mathbf{F}$.

Line integrals

3 Evaluate the line integral $\int_C (x + y)\, ds$ where C is the line segment from $(0, 0)$ to $(3, 4)$.

4 Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle y, x \rangle$ along $\mathbf{r}(t) = \langle t, t^2 \rangle$, $0 \le t \le 1$.

Conservative vector fields and potential functions

5 Determine whether $\mathbf{F}(x, y) = \langle 2xy, x^2 + 3y^2 \rangle$ is conservative, using the test $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.

6 Find a potential function $f(x, y)$ such that $\nabla f = \mathbf{F}$ for the field above.

The Fundamental Theorem for line integrals

7 Using the potential function $f(x, y) = x^2y + y^3$ found above, evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(0, 0)$ to $(1, 2)$, using the Fundamental Theorem for line integrals.

Green's Theorem

8 Use Green's Theorem, $\oint_C P\, dx + Q\, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$, to evaluate $\oint_C xy\, dx + x^2\, dy$ where C is the boundary of the square $0 \le x \le 1$, $0 \le y \le 1$.

9 Use Green's Theorem to find the area of the region enclosed by a curve C, using the formula $A = \frac{1}{2}\oint_C (x\, dy - y\, dx)$, for the circle $x = 2\cos t$, $y = 2\sin t$, $0 \le t \le 2\pi$.

10 Use Green's Theorem to evaluate $\oint_C (3y)\, dx + (5x)\, dy$ where C is the circle $x^2 + y^2 = 4$, traversed counterclockwise.

Work done by a force field

11 A force field $\mathbf{F}(x, y) = \langle y, x \rangle$ acts on an object moving along the straight path from $(0, 0)$ to $(2, 0)$ then from $(2, 0)$ to $(2, 3)$. Find the total work done, $W = \int_C \mathbf{F} \cdot d\mathbf{r}$, by evaluating each leg separately.

12 Confirm the work found above using the potential function $f(x, y) = xy$ (since $\mathbf{F} = \langle y, x \rangle = \nabla f$) and the Fundamental Theorem for line integrals.

Testing for conservative fields

13 Determine whether $\mathbf{F}(x, y) = \langle y^2, 2xy + 1 \rangle$ is conservative, and if so, find a potential function.

14 Determine whether $\mathbf{F}(x, y) = \langle 3x^2y, x^3 + 2y \rangle$ is conservative, and find a potential function if so.

Applying Green's Theorem to circulation

15 Use Green's Theorem to evaluate the circulation $\oint_C (-y)\, dx + x\, dy$ around the boundary of the rectangle $0 \leq x \leq 3$, $0 \leq y \leq 2$, traversed counterclockwise.

Line integrals over piecewise paths

16 Evaluate $\int_C y\, dx + x\, dy$ where C consists of the segment from $(0, 0)$ to $(1, 0)$ followed by the segment from $(1, 0)$ to $(1, 1)$.

17 Confirm the result above using the potential function $f(x, y) = xy$ (since $\langle y, x \rangle = \nabla f$) and the endpoints $(0, 0)$ and $(1, 1)$.

Flux and Green's Theorem (normal form)

18 Using the divergence form of Green's Theorem, $\oint_C \mathbf{F} \cdot \mathbf{n}\, ds = \iint_D \nabla \cdot \mathbf{F}\, dA$, find the outward flux of $\mathbf{F}(x, y) = \langle x, y \rangle$ across the boundary of the disk $x^2 + y^2 \leq 4$.

A capstone conservative-field application

19 A particle moves through a force field $\mathbf{F}(x, y) = \langle 2xy^3, 3x^2y^2 + 4 \rangle$ from the point $(0, 0)$ to $(2, 1)$ along any path. First verify that $\mathbf{F}$ is conservative, then find a potential function $f(x, y)$, and finally use it to compute the total work done by the field on the particle.

20 For the vector field $\mathbf{F}(x, y, z) = \langle yz, xz, xy + 2z \rangle$, complete each step:

a Find the divergence $\nabla \cdot \mathbf{F}$.

b Find the curl $\nabla \times \mathbf{F}$.

c Based on the curl, determine whether $\mathbf{F}$ could be conservative in 3D.